

# EvoRank: LLM-Guided Evolution of Multi-Objective Learning-to-Rank Pipelines

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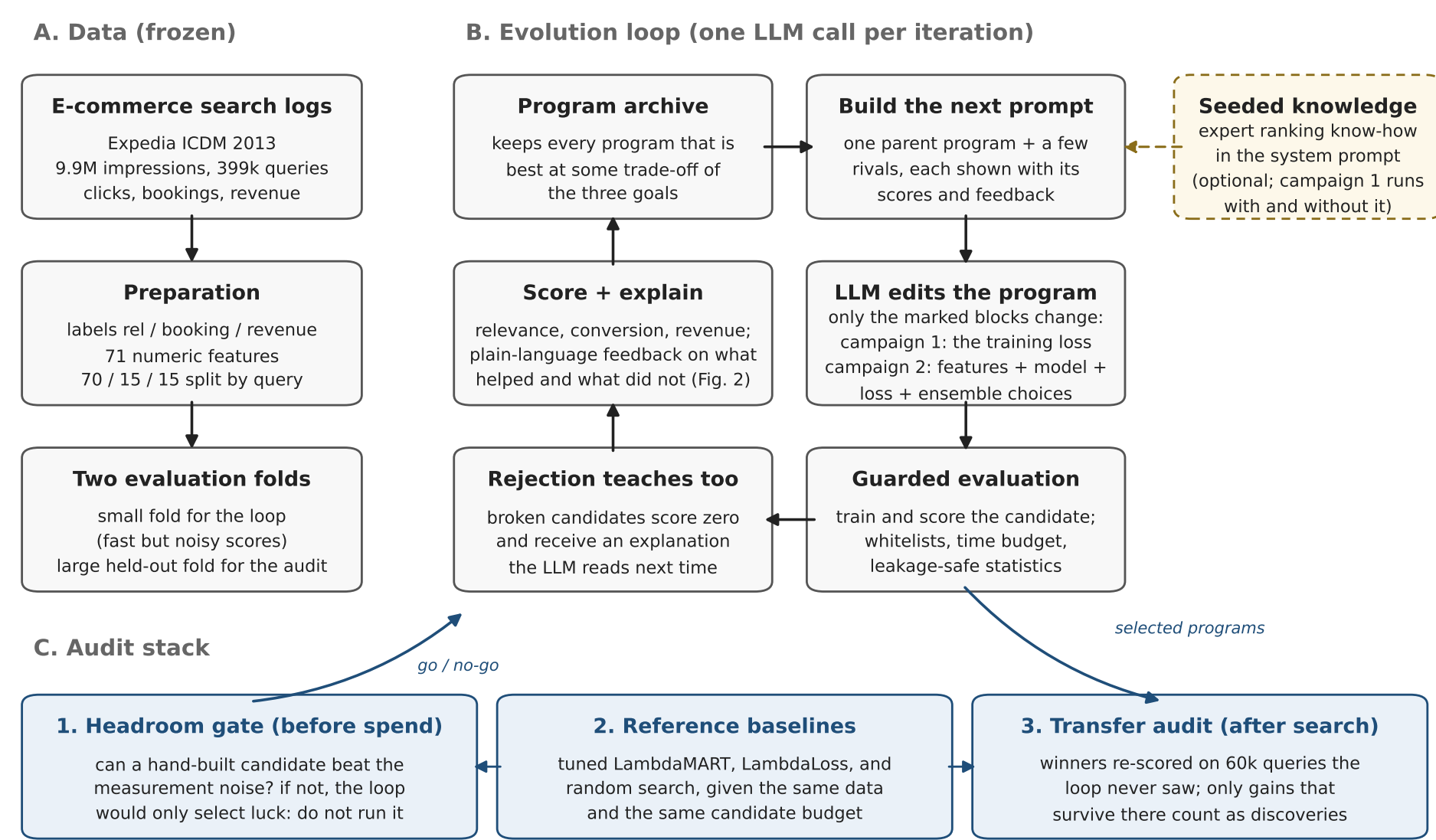
code + all results

## The failure nobody reports

LLM-guided discovery loops (AlphaEvolve-style) report gains on the data the loop used to select candidates. Ranking evaluation is noisy by necessity, and selection under noise promotes lucky candidates: a loop can show steady progress while discovering nothing that transfers.

**The claim.** Whether the loop pays off is measurable before any LLM spend: the ratio of achievable effect sizes in the search space to the noise of the fitness signal. We package this as the headroom gate, and let a transfer audit on held-out queries, not the loop’s own scores, decide what counts.

## One loop, fully audited



One LLM call per iteration edits the marked blocks of a candidate program; guarded evaluation scores it on a small fitness fold; the audit lane decides what is real. Engine: SkyDiscover (AdaEvolve controller).

The audit stack (acceptance and stopping criteria, identical in every campaign):

- (a) headroom gate: a hand-built candidate from the domain literature must beat the fitness fold’s noise floor by 2.5×
- (b) paired query bootstrap on every comparison
- (c) equal-budget random-search control
- (d) equal Optuna tuning on both sides
- (e) transfer audit: re-score every selected program on 60k held-out queries
- (f) hypervolume + component-removal tests

Task. Expedia hotel search (ICDM 2013): 9.9M impressions, 399k queries, split 70/15/15 by query. Three competing objectives from real behavior: relevance (NDCG@10), conversion (booking NDCG@10), revenue (pooled dollar-weighted capture).

## Campaign 1: selecting noise

Evolving the training objective (the LambdaMART gradient) looked great on the selection fold and failed the transfer audit, on 60k held-out queries:

- best evolved vs its own seed: **+0.0002** NDCG@10 ( $p = 0.76$ )
- blind random search transferred as well or better ( $p = 0.03$ )
- every method below tuned LambdaMART ( $p < 0.001$ )

Mechanism, measured: a single component’s true effect is **0.0005–0.008** NDCG@10; the fold’s measurement noise is  $\pm 0.007$ – $0.009$ . Picking the best score means picking luck.

Seeded domain knowledge made search faster, not truer: two of three seeded runs matched random search’s entire budget within 1–3 iterations (unseeded: 7–23), but what transferred was unchanged. Gradient-level diagnostic feedback changed nothing; enlarging the evaluation fold (1.6k→6k queries) halved noise and tripled surviving gain, the only intervention that moved it.

## Campaign 2: pipelines transfer

Same loop, wider space: features + model + loss + ensemble, with leakage-safe helpers (out-of-fold encodings). The gate passed first (**+0.013**  $\geq 2.5\times$  noise). Three runs, 50 iterations, ~\$10 each:

|                               | Val     | Test    | Test   | Test    |
|-------------------------------|---------|---------|--------|---------|
| Method (fitness-fold trained) | NDCG@10 | NDCG@10 | book   | revenue |
| Starting pipeline             | 0.4218  | 0.4222  | 0.4443 | 0.4105  |
| LambdaMART + Optuna           | 0.4330  | 0.4296  | 0.4527 | 0.4231  |
| Pipeline s0                   | 0.4356  | 0.4316  | 0.4545 | 0.4257  |
| Pipeline s1                   | 0.4359  | 0.4322  | 0.4553 | 0.4215  |
| Pipeline s2                   | 0.4398  | 0.4352  | 0.4582 | 0.4239  |

Transfer audit on 59,902 held-out queries; bold = column best.

Best run: **+0.0057** NDCG@10 over the tuned baseline (95% CI **[0.0041, 0.0073]**,  $p < 0.0001$ ), **+0.0056** booking NDCG@10, revenue matching; significantly better relevance and conversion at matching revenue, in every run.

## It holds at full scale

| 280k train / 60k test queries           | NDCG@10 | NDCG@38 |
|-----------------------------------------|---------|---------|
| Tuned baseline (small-fold tuning)      | 0.4349  | 0.4983  |
| Tuned baseline (full-scale tuning)      | 0.4544  | 0.5132  |
| Pipeline s1 (small-fold tuning)         | 0.4490  | 0.5080  |
| Pipeline s1 (baseline’s tuned settings) | 0.4619  | 0.5185  |

Features alone: 0.4587 (**+0.0043**); the ensemble adds **+0.0032**. Full-scale bootstrap: **+0.0075** NDCG@10,  $p < 0.0001$ .

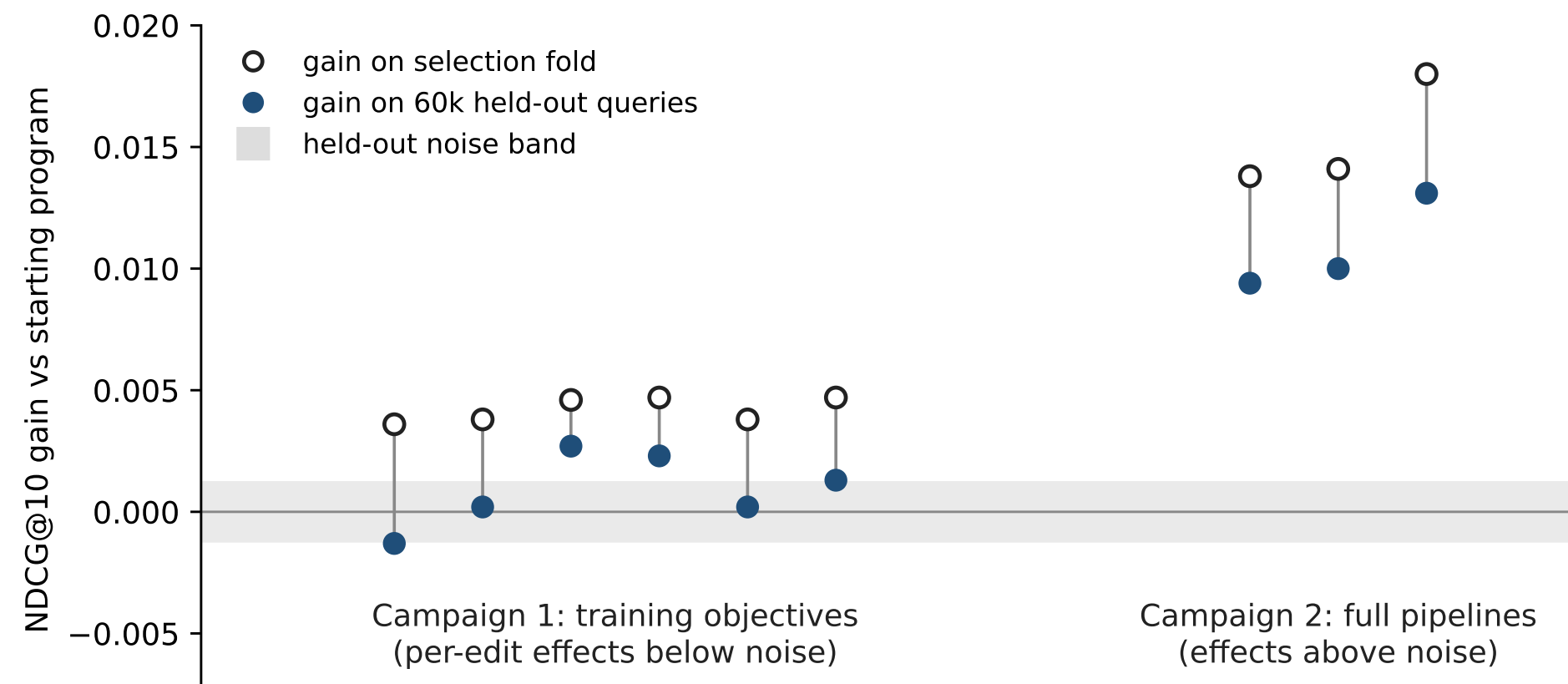
**Kaggle late submission** (one shot, hidden test set): discovered pipeline 0.5196 private NDCG@38 → would rank 20 of 340 (top 6%) in the 2013 challenge; tuned baseline 0.5140 (30th). Ten leaderboard places from the discovered structure alone.

## What the search found

```
for c in [price, stars, review, loc1, loc2]:
    out[c+"_qrnk"] = groupby(qid)[c].rank(pct=True)
out["prop_id_count"] = stats.count(df, "prop_id")
PIPELINE = {xgb:rankndcg + lgbm:lamdarank +
             sklearn:ert,
             ensemble: zscore_weighted [0.5, 0.35, 0.15]}
```

The same readable core in all three runs: within-query percentile ranks, count encodings, a lean feature set, and a diverse ensemble under per-query z-scores: the ICDM 2013 winners’ essential structure, rediscovered and extended autonomously.

## The study in one picture



Per-run NDCG@10 gains over the starting program, on the selection fold (open) vs 60k held-out queries (filled). Campaign 1 collapses into the noise band; campaign 2 survives.

Open everything: code, configs, seeds, logs, discovered programs, audit scripts at [github.com/shabazpatel/evorank](https://github.com/shabazpatel/evorank). Limitations: one dataset (the only public LTR set with conversion and revenue labels); three runs per condition; revenue is a proxy; audited for transfer, not online effects.

## Use this on your ranking stack

### 1. Measure noise

Paired query bootstrap on the evaluation fold you can afford: that half-width is your noise floor.

### 2. Hand-build one candidate

One reference candidate from your domain’s literature, before any LLM spend.

### 3. Gate the space

Require its gain  $\geq 2.5\times$  the noise floor. No headroom, no loop: keep your budget.

### 4. Run guarded

Leakage-safe statistics, whitelists, budgets, and rejection messages the model learns from.

### 5. Audit, then believe

A held-out transfer audit, never the loop’s own scores, decides what counts.